

Bitcoin Cryptocurrency

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1 BITCOIN CRYPTOCURRENCY

1.1 Bitcoin Transactions and Maintaining Blockchains

Bitcoin is essentially a purely digital currency with no central control from any type of government or organization. The system relies completely on peer to peer software for all transactions and blockchain interactions. The bitcoin system utilizes crowd computing to be able to work similarly to what a physical wallet would for the bitcoin digital wallet. All transactions involving bitcoin are recorded in a public ledger that is held on servers that any user can set up. A user must join the Bitcoin network to be given a public and private key for the blockchain. Each access to the network will work as a node that is then used to broadcast every transaction from node to node. These transactions are then compiled into a block which is permanently added to the blockchain. The bitcoin system basically uses each user in the network as a node to broadcast transactions involving bitcoin to each other, which essentially is simply using crowd computing for transaction management. In addition, the system also utilizes the same process in maintaining blockchains as well.

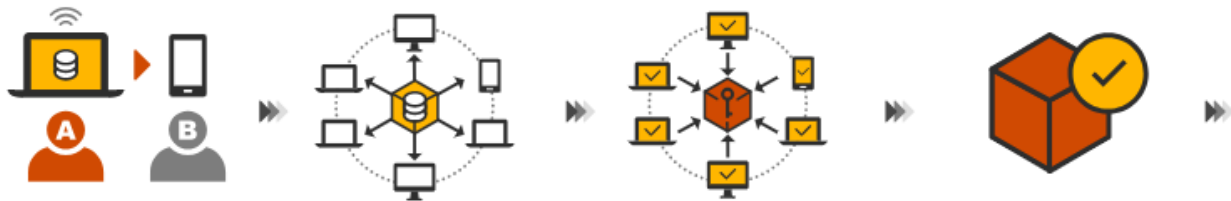


Figure 1—Process of Blockchain Part 1. [Source](#)



Figure 2—Process of Blockchain Part 2. [Source](#)

The figure displays that once a transaction occurs, the transaction is broadcasted throughout the p2p network. Each node will receive notification of the transaction. Once the transaction occurs then the ledger for all the transactions are compiled together into a block and added towards the blockchain and the transaction is complete. The p2p network and crowd computing are two aspects of this entire process that helps maintain the blockchain itself.

1.2 Digital Signature

Every user in the bitcoin network is given a public key and a private key. The public key is used as an identification tool for the user's computer. The private key on the other hand should be known solely by the user themselves. These keys are used together in the process of creating and verifying a digital signature that a bitcoin transaction can have.

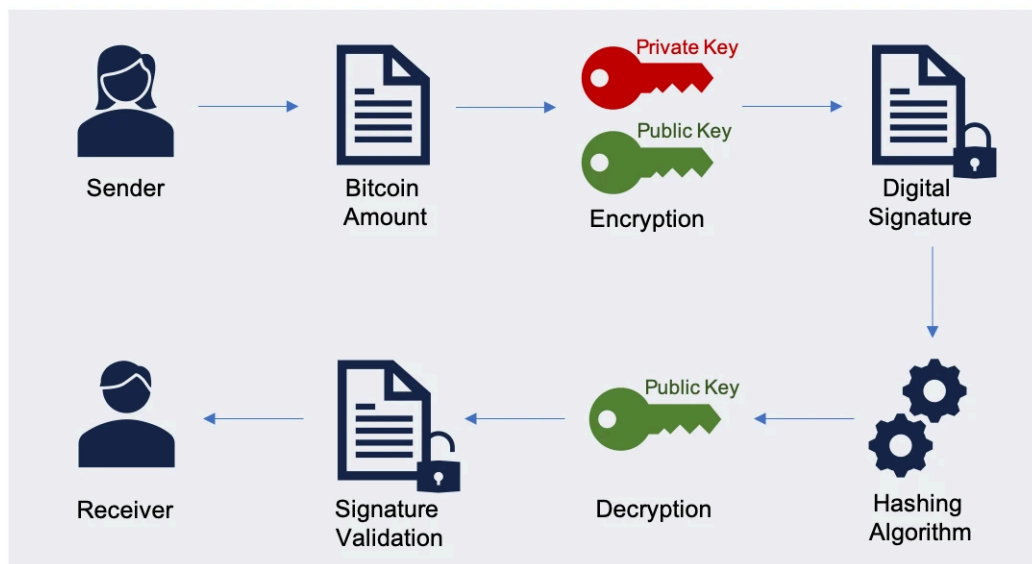


Figure 3—Process of Bitcoin Digital Signature. [Source](#)

Whenever a bitcoin transaction occurs the user of the transaction is able to take his or her private key to sign a message that he or she has written that has been encrypted using the receiver's public key. The encrypted message is then combined with the digital signature to become the transaction to the receiver. Each and every one of these transactions are broadcast throughout the peer to peer network within the bitcoin network. The transaction is then compiled into a block to be permanently added to the blockchain. This ensures that every single bitcoin that flows through from one account to another is not only broadcast throughout the network but is also recorded on the blockchain permanently. This process ultimately is able to keep track of every bitcoin every user has, but also who has the bitcoin before as well.

1.3 Blockchain Technology Applications

Blockchain technology is an extremely useful technology that can be extended past cryptocurrency into other parts of everyday life. The most relevant application currently, in my opinion, is voting. With the presidential election currently going on, blockchain technology can simplify the process of voting for the entire country. Since the blockchain can be replicated to mimic the voting process as an individual transaction, this automatically allows each person to be able to vote without having to physically go vote in person. In addition, with the blockchain's integrity checks, the process of ensuring that each person voting can only be voting once would be simple since he or she would be required to use some form of authentication such as SSN. Another aspect of society that blockchain could benefit would be the implementation for individual identification in blockchain. Each person hypothetically could be represented by a bitcoin and his or her identification would be used together with his or her SSN to create a digital signature for each interaction. This would allow the government to be able to move each bitcoin as a person so that separation and identification would be much smoother and there would be a complete ledger of how each person is separated. This would speed up processes such as disability applications, driver's license, background checks, etc. The last application I believe would benefit from the inclusion of block chain technology would be real estate. Currently, the process of buying a home is extremely convoluted and unnecessarily complicated. By using block chain technology, homes would be able to be represented with bitcoins and the transaction history would be permanently within the blockchain along with all the information of the home as well. This could essentially assist home buyers and owners in all the research and paperwork of buying homes.

2 BLOCKCHAIN

2.1 Digital Signature

The digital signature used within every bitcoin transaction mainly has the job of ensuring the integrity of the bitcoin. The blockchain has records of every single bitcoin transaction that has occurred along with the digital signature of each transaction. To ensure integrity of the entire process, only verified bitcoin transactions can be added to the blockchain. In addition, once a block has been added to the chain, it is extremely difficult to remove or change any of the data in the blockchain. This means that transactions within the blockchain need to be verified before it can be added to the blockchain to ensure everything within the chain is correct. Please take a look at Figure 2. The verification process of the transaction includes taking the digital signature of the transaction and the public key of the sender's transaction to check if the user is really the one who sent the transaction. The message will be taken and hashed which will then be compared to the unencrypted hash with the sender's public key. As long as the values match the

verification is correct and the transaction can be completed as normal and added to the blockchain. This entire process makes it extremely difficult for any problems to arise during bitcoin transactions and the integrity of these transactions can be ensured.

2.2 Double Spending of Bitcoin

Double spending is the issue in which the same cryptocurrency is used more than once. In this situation, the bitcoin system utilizes a selection of different techniques to counteract the issue of double spending. The main tool used to resolve this issue is the timestamping of transactions so that a chronological list of transactions are able to verify if a transaction is double spent. The peer-to-peer network used for bitcoin uses the timestamp so that all the nodes in the network must have consensus for the transaction to be added onto the blockchain. In addition, the consensus algorithm PoW requires nodes to generate computational proof through problems to create a chronological order for the transactions. This will ensure that the transactions are time stamped and the bitcoin cannot be double spent since there is an order.

2.3 Privacy Key

A user's public key and private key are extremely important and should never be lost. However, in this hypothetical situation, if Alice lost her privacy key, there is essentially no possible way for her to be able to recover her key. The system is specifically designed so that there is no way to recover the key. This is set up as such so that criminals are not able to use recovery methods to steal a user's key. Personally, I believe there should not be measures set up to allow a user to recover the key. However, if this does happen to a user, I believe that he or she should not just lose access to all of his or her cryptocurrency. I would suggest implementing a system that allows the user to liquidate his or her cryptocurrency after a set amount of time that the user has not interacted with the cryptocurrency. This can be setup by the user when he or she is creating the account. Since the user sets up this option, once a certain amount of time the user has not touched his or her cryptocurrency, they would be given the option of liquidation without having to use their private key. This will allow users to retrieve their money instead of losing it all whenever they lose their private key.

3 CRYPTOCURRENCY OPTIMIZATION

3.1 P2P Network in Blockchain

Bitcoin's system is able to ensure the availability of the p2p network to make a bitcoin transaction at any time through the use of the proof of work consensus mechanism and the replicated ledger. Whenever a transaction has begun the bitcoin transaction is broadcast across the peer to peer network so that multiple nodes receive the message. The nodes would then check whether or not the transaction is valid which would then have the PoW consensus mechanism

have miners compete to solve computational mathematical problems to create a block to add to the block chain. With multiple nodes going through this process at the same time, the ledger would need to have consensus to have the block added to the block chain. This essentially ensures that the peer to peer network will always have the availability to create a bitcoin transaction at any time.

3.2 Confidentiality

Bitcoin's system is able to ensure confidentiality of its data through the strategic use of pseudonymity. Although there is no anonymity within bitcoins, there is no possible way for a user's sensitive information to be leaked. A user is only given a public key and private key. Public keys are already known to the public and there is no sensitive data possible to be found within public keys. The user's private key is only known to the user and is used for creating the digital signature. The digital signature then uses the user's public key for decrypting, so there is no sensitive data either. Throughout the whole process of a bitcoin transaction, all the data that is broadcast and that can be known are only information that identifies a transaction that is ongoing and the user through the public key. However, the pseudonymity through the digital signature and public key ensures that users have pseudonymity within the bitcoin system.

3.3 Bitcoin Privacy

Although the bitcoin system ensures the user's confidentiality through pseudonymity, there are still possible ways for the user's sensitive information to be revealed. In some situations, user information can be extracted from context outside of the bitcoin transaction. If a business or organization you are sending bitcoin to and from collects personal client information then they would be able to identify the user through the bitcoin transaction. Bitcoin does not provide privacy beyond pseudonymity since the blockchain and bitcoin transactions are public. Similarly, Ethereum also can only provide the users pseudonymity and nothing beyond that.

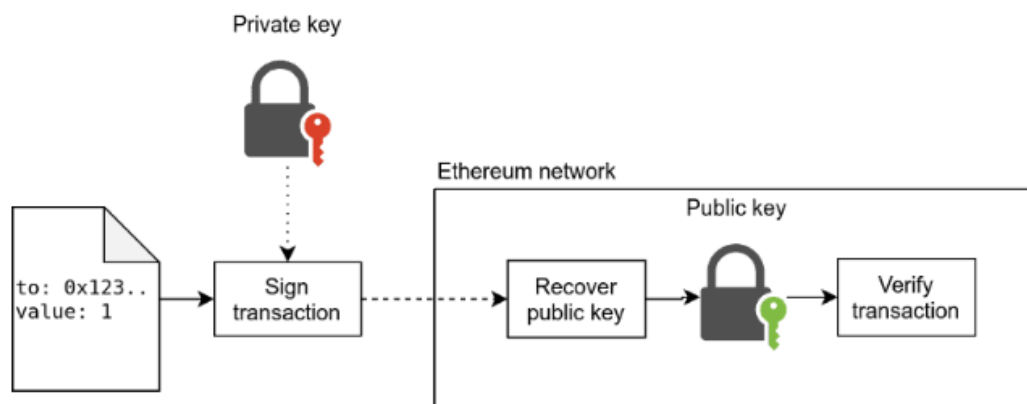


Figure 4—Process of Ethereum Transaction. [Source](#)

Ethereum's transactions are also public since the blockchain is public and anyone is able to view the transactions within. The transactions of Ethereum also do not involve any kind of sensitive information from the user and use the user's public and private key to keep identities from being revealed.

4 TRUST MANAGEMENT

4.1 Pros/Cons

Institutional and social are two different approaches in which trust behavior is encouraged in a decentralized peer to peer system. An institutional approach has agents that are counted as a part of the institution. Within the institution the agents are then either rewarded or punished based on his or her behavior. With this type of approach towards the agents, these people will have more of an incentive towards higher trust behavior. They are both rewarded and punished based on their interactions so the agents will have more reason to behave trustworthy. Although this encourages higher trust behavior this is completely proportional to how well the rewards and punishments are, which if the agent is indifferent to both can facilitate malicious behavior instead. A social approach involves having agents evaluate others through reputation and exchange information with others as well. This gives agents a clear visual of increase and decrease in reputation since higher or lower reputation can affect how others in society interact with the agent. This gives agents a very high incentive to behave trustworthy to increase reputation. Although this can also be utilized negatively if an agent behaves maliciously to purposely decrease another person's reputation. In this approach the positive and negative are clearly shown to others which gives everyone incentive for higher trust behavior.

4.2 Feedback Rating

Trust rating systems today require feedback rating only after legitimate transactions as a defense mechanism against the manipulation of service experience. The trust developed from a user reading the rating of a service can be directly affected if the rating of a service is too low. To decrease the chances of people purposely lowering or raising the ratings of a service, the rating systems today can only be done after a legitimate transaction is completed. This ensures that people who have not received the service cannot manipulate the ratings in any way they want. In addition, it also ensures that only people who have received the service can rate the service. Although this helps the accuracy of the ratings of the service, there are still possible ways around this system. As long as the user has completed a legitimate transaction then manipulation of the ratings is still possible. Competitors or other people who are incentivized enough to purchase the service will be able to enter a review and rating however they would like. Similarly, people who support the service will be able to do the same as well. Although having the feedback rating

available only after transactions deter people from rating manipulation, if someone is willing enough there are ways around it.

4.3 Eigen Trust

Eigen Trust is insufficient for trust propagation against malicious individual groups since the system itself will not be able to handle the false information provided by the malicious nodes. These nodes will be able to offer false information that can lead to an increase in trust in the malicious nodes. Since Eigen Trust relies on peer to peer trust the evaluation process will be completely falsified since there is no way to verify the trust of the nodes from other nodes. In addition, the malicious nodes could work together to manipulate the trust on specific nodes to lose trust. Eigen Trust ultimately is not effective with trust propagation when dealing with malicious individuals.

5 REFERENCES

1. Sparkes, M. (2021). *What is bitcoin and how does it work?* New Scientist. <https://www.newscientist.com/definition/bitcoin/>
2. SoluLab. (2021, August 9). *SoluLab - Blockchain Development Company*. Blockchain Technology, Mobility, AI and IoT Development Company USA, Canada. <https://www.solulab.com/how-blockchain-anonymity-is-maintained/>